

A Complete Low-Exponent Anisotropic Grothendieck Coincidence Theorem and a High-Exponent Hadamard Obstruction

Automated mathematics run `fa_banach_001`, lane 2

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Status: substantial partial result likely valid. For arbitrary anisotropic input exponents in $[1, 2]$, we give a complete if-and-only-if classification. The source's unrestricted high-exponent classification remains open, but a new ordered pairwise obstruction rules out the naive coordinatewise extension.

Abstract

Albuquerque, Araújo, Rezende, and Santos ask for which anisotropic multi-indices every bounded m -linear map $\ell_1^m \rightarrow \ell_2$ is multiple summing. We completely answer the question when all input exponents lie in $[1, 2]$: for $\mathbf{p} = (p_1, \dots, p_m) \in [1, 2]^m$ and $\mathbf{q} = (q_1, \dots, q_m) \in [1, \infty)^m$, the coincidence holds if and only if $q_k \geq p_k$ for every k . At the vertices $p_k \in \{1, 2\}$, mixed-norm Minkowski reorders the exponent-one levels outside the exponent-two levels. The inner block is controlled by the multilinear $(2; 2)$ Grothendieck theorem and the outer block by the $(1; 1)$ theorem, both with Hilbert-valued range. Coordinatewise complex interpolation fills the cube $[1, 2]^m$, and mixed-norm monotonicity gives all $q_k \geq p_k$. Rank-one maps prove necessity. Beyond this cube, a Hadamard construction shows that every coincidence tuple must satisfy

$$\frac{1}{q_k} \leq \frac{1}{p_\ell} + \left(\frac{1}{p_k} - \frac{1}{2} \right)_+ \quad (k < \ell).$$

Thus the unrestricted problem has genuinely ordered cross-coordinate constraints; coordinatewise scalar conditions cannot classify it.

1 Source question

The source is N. Albuquerque, G. Araújo, L. Rezende, and J. Santos, *A summability principle and applications*, arXiv:1904.04549v2 (2025)[1]. On PDF page 9 it asks for which anisotropic multi-indices $(\mathbf{q}; \mathbf{p})$ one has

$$\Pi_{(\mathbf{q}; \mathbf{p})}^{\text{ms}}({}^m\ell_1; \ell_2) = \mathcal{L}({}^m\ell_1; \ell_2).$$

The source recalls the complete isotropic answer and supplies additional sufficient anisotropic families, but not a full classification.

For Banach-space-valued arrays $a = (a_{i_1, \dots, i_m})$, write

$$\|a\|_{\ell_{\mathbf{q}}} := \left(\sum_{i_1} \left(\cdots \left(\sum_{i_m} \|a_{i_1, \dots, i_m}\|^{q_m} \right)^{q_{m-1}/q_m} \cdots \right)^{q_1/q_2} \right)^{1/q_1}.$$

Corollary 4.4 recovers [5, Corollary 2] and, when $d = 1$, recovers a result of [32]. The following example is intended to illustrate this.

Example 4.5. Suppose $m = 3, d = 2, p = 4, I_1 = \{1, 2\}$ and $I_2 = \{3\}$. By Theorem 4.2 we know that (4.1) holds with $s_k = 4$ for $k = 1, 2$, whereas by Theorem 4.3 we have $s_1 = 4$ and $s_2 = 12/5$.

4.2. A new Grothendieck inclusion result. Grothendieck’s famous theorem for absolutely summing linear operators has been extended to the m -linear setting (see [13, Theorems 5.1 and 5.2] and [26]). More specifically, every continuous m -linear operator $T: \ell_1 \times \cdots \times \ell_1 \rightarrow \ell_2$ is multiple p -summing for all $p \in [1, 2]$. A natural question arises: for which pairs (q, p) does the coincidence

$$\Pi_{(q,p)}^{\text{ms}}({}^m\ell_1; \ell_2) = \mathcal{L}({}^m\ell_1; \ell_2)$$

hold? In [25] the authors present a definitive answer for this question.

Theorem 4.6. [25] Let m be a positive integer and $1 \leq p \leq q < \infty$. Then $\Pi_{(q,p)}^{\text{ms}}({}^m\ell_1; \ell_2) = \mathcal{L}({}^m\ell_1; \ell_2)$ if and only if $p \leq 2$ or $q > p > 2$.

Given the various generalizations of this theory to anisotropic settings, a natural question arises: For which multi-indices $(\mathbf{q}; \mathbf{p}) = (q_1, \dots, q_m; p_1, \dots, p_m)$, with $q_i \neq q_j$ or $p_i \neq p_j$ for some $i, j \in \{1, \dots, m\}$, does the equality

$$\Pi_{(\mathbf{q}; \mathbf{p})}^{\text{ms}}({}^m\ell_1; \ell_2) = \mathcal{L}({}^m\ell_1; \ell_2)$$

hold? In [11, Proposition 4.1(a)], it was shown that $\Pi_{(q; 1, \dots, 1, q)}^{\text{ms}}({}^m\ell_1; \ell_2) = \mathcal{L}({}^m\ell_1; \ell_2)$ for all $q \geq 2$. As a consequence of Theorem 3.2, we provide two additional solutions to this problem.

Proposition 4.7. Let m be a positive integer, $p \leq 2$ or $r > p > 2$ and $\mathbf{s}, \mathbf{q} \in [1, \infty)^m$. If $1/r - m/p + |1/\mathbf{q}| > 0$, and $q_k \geq p$ and

$$\frac{1}{s_k} - \left| \frac{1}{\mathbf{q}} \right|_{\geq k} = \frac{1}{r} - \frac{m - k + 1}{\mathbf{p}}$$

for $k = 1, \dots, m$, then

$$\Pi_{(\mathbf{s}; \mathbf{q})}^{\text{ms}}({}^m\ell_1; \ell_2) = \mathcal{L}({}^m\ell_1; \ell_2).$$

Figure 1: The anisotropic classification question and the source’s Proposition 4.7, PDF page 9.

An m -linear $T: \ell_1^m \rightarrow \ell_2$ is multiple $(\mathbf{q}; \mathbf{p})$ -summing when

$$\|(T(x_{i_1}^1, \dots, x_{i_m}^m))_{\mathbf{i}}\|_{\ell_{\mathbf{q}}} \leq C \prod_{k=1}^m \|(x_i^k)_i\|_{w, p_k} \quad (1)$$

for every finitely supported choice of sequences.

2 Proof intuition

The isotropic theorem gives the two endpoints needed for each coordinate: every m -linear map $\ell_1^m \rightarrow H$, with H Hilbert, is both multiple $(1; 1)$ - and multiple $(2; 2)$ -summing. A mixed vertex such as $(1, 2, 1, 2)$ appears different, but Minkowski’s inequality moves the smaller exponents outward, where the mixed norm is largest. After this reordering, the exponent-two coordinates form one Hilbert-valued inner block. The exponent-one coordinates then act on that Hilbert space as an outer block. Applying the endpoint theorem twice proves every $\{1, 2\}$ -vertex.

The remaining step is genuinely anisotropic: interpolate one coordinate at a time, simultaneously interpolating its weak sequence space and the corresponding level of the mixed output norm. The required uniform identity for weak sequence spaces over ℓ_1 is the standard injective-tensor interpolation formula used in Bayart–Pellegriño–Rueda[2]. This fills the entire low-exponent cube.

3 The exact low-exponent theorem

Theorem 1. *Let $m \geq 1$, let H be a Hilbert space, and let*

$$\mathbf{p} = (p_1, \dots, p_m) \in [1, 2]^m, \quad \mathbf{q} = (q_1, \dots, q_m) \in [1, \infty)^m.$$

Then

$$\Pi_{(\mathbf{q}; \mathbf{p})}^{\text{ms}}({}^m\ell_1; H) = \mathcal{L}({}^m\ell_1; H)$$

if and only if $q_k \geq p_k$ for every k . In particular, this completely classifies the source problem throughout the input cube $[1, 2]^m$.

Proof. We first prove sufficiency when $q_k = p_k \in \{1, 2\}$ for every k . For a nonnegative matrix and $1 \leq r \leq s$, Minkowski's inequality gives

$$\left(\sum_i \left(\sum_j a_{ij}^r \right)^{s/r} \right)^{1/s} \leq \left(\sum_j \left(\sum_i a_{ij}^s \right)^{r/s} \right)^{1/r}. \quad (2)$$

Thus each adjacent pattern with exponent 2 outside exponent 1 is bounded by the transposed pattern with exponent 1 outside exponent 2. Repeatedly applying (2), while permuting the corresponding variables of T , reduces every vertex to

$$\underbrace{(1, \dots, 1)}_{a \text{ times}}, \underbrace{(2, \dots, 2)}_{b \text{ times}}, \quad a + b = m.$$

Fix sequences in the last b variables. Define

$$S(x_1, \dots, x_a) = (T(x_1, \dots, x_a, x_{a+1}^{a+1}, \dots, x_{i_m}^m))_{i_{a+1}, \dots, i_m}.$$

The target $\ell_2(H)$ over the last b indices is Hilbert. The multilinear $(2; 2)$ Grothendieck coincidence, which is the $p = 2$ case of source Theorem 4.6, gives

$$\|S\| \leq C_{2,b} \|T\| \prod_{k=a+1}^m \|(x_i^k)_i\|_{w,2}.$$

Now apply the $(1; 1)$ coincidence to the a -linear map S , again with Hilbert target. We obtain

$$\begin{aligned} & \sum_{i_1, \dots, i_a} \left(\sum_{i_{a+1}, \dots, i_m} \|T(x_{i_1}^1, \dots, x_{i_m}^m)\|^2 \right)^{1/2} \\ & \leq C_{1,a} C_{2,b} \|T\| \prod_{k=1}^a \|(x_i^k)_i\|_{w,1} \prod_{k=a+1}^m \|(x_i^k)_i\|_{w,2}. \end{aligned}$$

This is exactly the sorted mixed-vertex estimate. The cases $a = 0$ or $b = 0$ use only the corresponding endpoint theorem.

We next fill the cube. For a finite sequence of length N , put

$$W_p^N = \ell_p^{w,N}(\ell_1).$$

The canonical identification

$$W_p^N = \ell_p^N \widehat{\otimes}_\varepsilon \ell_1$$

is isometric. Uniformly in N , injective-tensor interpolation gives an isomorphism

$$[W_1^N, W_2^N]_\theta \simeq W_p^N, \quad \frac{1}{p} = 1 - \frac{\theta}{2}. \quad (3)$$

This is the ℓ_1 weak-sequence interpolation formula used in [2, Section 2]; it follows from the injective interpolation theorem there. Vector-valued sequence-space interpolation also gives, at any fixed level of an iterated mixed norm,

$$[\ell_1(Y), \ell_2(Y)]_\theta = \ell_p(Y).$$

For fixed T and N , its coefficient map is multilinear from the product of the $W_{p_k}^N$ spaces to the corresponding finite mixed sequence space. We have just proved a uniform bound at all 2^m cube vertices. Apply multilinear complex interpolation first in coordinate 1, then coordinate 2, and so on, using (3) and the vector-valued output formula at each step. This proves (1) with $\mathbf{q} = \mathbf{p}$ for every $\mathbf{p} \in [1, 2]^m$, with a constant independent of N . Taking the supremum over finite truncations gives the infinite-sequence assertion. For real scalars, complexification gives the same coincidence with a harmless constant depending only on m .

If $q_k \geq p_k$ for every k , coordinatewise monotonicity of iterated ℓ_p norms yields $\|a\|_{\ell_{\mathbf{q}}} \leq \|a\|_{\ell_{\mathbf{p}}}$, proving sufficiency in full.

For necessity, fix k . Choose norm-one functionals $\varphi_j \in \ell_1^* = \ell_\infty$, a unit vector $h \in H$, and the rank-one map

$$T(x_1, \dots, x_m) = \varphi_1(x_1) \cdots \varphi_m(x_m) h.$$

In every coordinate except k , use a one-term sequence on which φ_j equals one. In coordinate k , use $x_i = a_i u$ with $\varphi_k(u) = 1$. Inequality (1) reduces to

$$\|(a_i)\|_{q_k} \leq C \|(a_i)\|_{p_k}.$$

Uniformity over finite scalar sequences forces $q_k \geq p_k$. This holds for every k , completing the proof. $\square$

4 A high-exponent Hadamard obstruction

The low-exponent theorem does not extend coordinatewise once a later input exponent is allowed above two. The following necessary condition makes the new obstruction explicit.

Proposition 2. *Assume $m \geq 2$ and*

$$\Pi_{(\mathbf{q}; \mathbf{p})}^{\text{ms}}({}^m \ell_1; H) = \mathcal{L}({}^m \ell_1; H)$$

for a nonzero Hilbert space H . Then, besides $q_j \geq p_j$ for every j , one necessarily has

$$\frac{1}{q_k} \leq \frac{1}{p_\ell} + \max \left\{ \frac{1}{p_k} - \frac{1}{2}, 0 \right\} \quad \text{for every } 1 \leq k < \ell \leq m. \quad (4)$$

Proof. Equality of the two operator spaces and the open mapping theorem give a constant C , independent of T , in (1). Fix $k < \ell$ and let N run through powers of two. Let $A_N = (\varepsilon_{ab})_{a,b=1}^N$ be a Sylvester Hadamard matrix, so $A_N A_N^T = N I_N$. On the k -th copy of ℓ_1 , put

$$x_i^k = \frac{1}{N} \sum_{a=1}^N \varepsilon_{ia} e_a, \quad i = 1, \dots, N,$$

and on the ℓ -th copy put $x_j^\ell = e_j$. For every other coordinate use a single unit vector.

If $z \in B_{\ell_\infty^N}$, orthogonality gives $\|A_N z\|_2 = \sqrt{N}\|z\|_2 \leq N$. Comparing finite-dimensional ℓ_p -norms therefore yields

$$\|(x_i^k)_{i=1}^N\|_{w,p_k} = \frac{1}{N} \|A_N : \ell_\infty^N \rightarrow \ell_{p_k}^N\| \leq N^{(1/p_k - 1/2)_+}. \quad (5)$$

Also $\|(e_j)_{j=1}^N\|_{w,p_\ell} = N^{1/p_\ell}$.

Choose a unit vector $h \in H$, norm-one functionals in the unused variables, and define T_N by inserting in coordinates k, ℓ the bilinear form

$$B_N(u, v) = \sum_{a,b=1}^N \varepsilon_{ab} u_a v_b$$

and multiplying by h and those functionals. Since the domain is a product of ℓ_1 -spaces, $\|T_N\| = \max_{a,b} |\varepsilon_{ab}| = 1$. Hadamard orthogonality gives

$$T_N(x_i^k, e_j) = \delta_{ij} h.$$

All intervening indices have length one, so the mixed norm of this diagonal array is N^{1/q_k} . Applying (1) and (5) gives

$$N^{1/q_k} \leq C N^{(1/p_k - 1/2)_+ + 1/p_\ell}.$$

Letting $N \rightarrow \infty$ proves (4). $\square$

For example, $\mathbf{p} = (3, 100)$ and $\mathbf{q} = (4, 101)$ satisfy the naive coordinatewise inequalities, including strict improvement in both high coordinates, but Proposition 2 would require $1/4 \leq 1/100$. Hence this tuple is not a coincidence tuple.

5 Verification, novelty, and remaining range

The script `code/verify_mixed_norm_reductions.py` checks the adjacent Minkowski direction, bubble-sort reduction of arbitrary $\{1, 2\}$ -patterns, and coordinatewise mixed-norm monotonicity on random arrays in dimensions two through five. It also checks the Sylvester orthogonality and weak-norm estimate used in Proposition 2. These computations guard against norm-order and scaling mistakes; they are not part of the proof.

The theorem is sharp and complete for $p_k \leq 2$. The unrestricted source question remains open here. Freezing all other variables gives only the coordinatewise necessity $q_k \geq p_k$. Proposition 2 shows that this is far from sufficient: later input exponents constrain earlier output levels. A complete sufficiency theorem for these ordered high-exponent interactions is not supplied by the known isotropic theorem or the source’s suffix-balance inclusion families.

A bounded novelty audit on 26 August 2026 searched all four run indexes, the local parsed arXiv corpus, and arXiv/web results using the source id/title and the phrases “anisotropic Grothendieck,” “mixed norm multiple summing,” and “ ℓ_1 to ℓ_2 anisotropic coincidence.” It found the source, the earlier isotropic and interpolation papers, and no statement of the complete $[1, 2]^m$ classification above. The source’s 2025 revision still poses the general question. This is bounded evidence, not a priority claim.

Human review recommendation: **send to human**, focusing on the coordinatewise interpolation step, uniformity of (3), and the exponent direction in (4).

References

- [1] N. Albuquerque, G. Araújo, L. Rezende, and J. Santos, *A summability principle and applications*, arXiv:1904.04549v2, 2025.
- [2] F. Bayart, D. Pellegrino, and P. Rueda, *On coincidence results for summing multilinear operators: interpolation, ℓ_1 -spaces and cotype*, arXiv:1805.12500, 2018.